

An Intuitive Proof to Central Limit Theorem

- ξ_i i.i.d. with $\mathbb{E}\xi_i = 0, \mathbb{E}\xi_i^2 = 1$. Then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i \xrightarrow{d} N(0, 1)$$

proof

It suffices to prove for any smooth and compactly supported f , we have

$$\lim_{n \rightarrow \infty} \mathbb{E}f\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i\right) = \mathbb{E}f(Z)$$

where $Z \sim N(0, 1)$.

Write Z as

$$Z = \frac{1}{\sqrt{n}} \sum_{i=1}^n \zeta_i$$

where $\zeta_i \stackrel{\text{i.i.d}}{\sim} N(0, 1)$.

Take $\zeta'_i = \frac{1}{\sqrt{n}} \zeta_i, \xi'_i = \frac{1}{\sqrt{n}} \xi_i$.

We have

$$\begin{aligned} & \left| \mathbb{E}f\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i\right) - \mathbb{E}f\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \zeta_i\right) \right| \\ & \leq \sum_{k=1}^n \left| \mathbb{E}f\left(\sum_{i=1}^k \xi'_i + \sum_{i=k+1}^n \zeta'_i\right) - \mathbb{E}f\left(\sum_{i=1}^{k-1} \xi'_i + \sum_{i=k}^n \zeta'_i\right) \right| \end{aligned}$$

By Taylor's expansion,

$$\left| f(x+y) - f(x) - f'(x)y - \frac{1}{2}f''(x)y^2 \right| \leq \begin{cases} \varepsilon y^2, & |y| \leq \delta \\ Cy^2, & \text{else} \end{cases}$$

Then

$$\begin{aligned} & \mathbb{E}f\left(\sum_{i=1}^k \xi'_i + \sum_{i=k+1}^n \zeta'_i\right) - \mathbb{E}f\left(\sum_{i=1}^{k-1} \xi'_i + \sum_{i=k}^n \zeta'_i\right) \\ & = \mathbb{E} \left\{ f' \left(\sum_{i=1}^{k-1} \xi'_i + \sum_{i=k}^n \zeta'_i \right) (\xi'_k - \zeta'_k) + \frac{1}{2} f'' \left(\sum_{i=1}^{k-1} \xi'_i + \sum_{i=k}^n \zeta'_i \right) (\xi'^2_k - \zeta'^2_k) \right\} + \mathbb{E}R_k^{(1)} + \mathbb{E}R_k^{(2)} \end{aligned}$$

The first term equals to 0 due to independence. And $R_k^{(i)}, i = 1, 2$ are two residuals.

We then have

$$\begin{aligned} \left| \mathbb{E}R_k^{(1)} + \mathbb{E}R_k^{(2)} \right| & \leq \varepsilon \mathbb{E}\xi'^2_k + \overline{C \mathbb{E}\xi'^2_k 1_{\{|\xi'_k| > \delta\}}} + \varepsilon \mathbb{E}\zeta'^2_k + C \mathbb{E}\zeta'^2_k 1_{\{|\zeta'_k| > \delta\}} \\ & \leq 2\frac{\varepsilon}{n} + \frac{C}{n} \mathbb{E}\xi_k^2 1_{\{|\xi_k| > \sqrt{n}\delta\}} + \frac{C}{n} \mathbb{E}\zeta_k^2 1_{\{|\zeta_k| > \sqrt{n}\delta\}} \end{aligned}$$

Sum over k from 1 to n , we get

$$\begin{aligned} & \left| \mathbb{E}f\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i\right) - \mathbb{E}f\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \zeta_i\right) \right| \\ & \leq 2\varepsilon + C\mathbb{E}\xi^2 1_{\{|\xi| > \sqrt{n}\delta\}} + C\mathbb{E}\zeta^2 1_{\{|\zeta| > \sqrt{n}\delta\}} \end{aligned}$$

Let $n \rightarrow \infty$, by DCT, we get

$$\limsup_{n \rightarrow \infty} \left| \mathbb{E}f\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i\right) - \mathbb{E}f\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n \zeta_i\right) \right| \leq 2\varepsilon$$

Then let $\varepsilon \rightarrow 0^+$, we complete the proof.

Comment: From this proof, we can derive the Lindeberg condition from the underlined expression.